

Builder Journal

Hand Gesture Detection — Week 2, AI Summer Fellowship 2026

Dataset Challenges

Reaching the 311-image target was actually the easier part — once I had a clear per-class count in mind, collecting the raw volume didn't take too long. The harder part was variety, not quantity. Getting different angles, distances, and hand orientations for each gesture took much more effort than just taking more photos of the same pose. Early on, most of my images were shot from similar angles, which meant the dataset looked bigger than it actually was in terms of what the model could learn from.

Annotation Challenges

My first annotation pass had a real problem: the bounding boxes were too big — often including wrist, forearm, or background space around the hand instead of being tight around just the gesture itself. This taught the model the wrong region mattered for each class. I went back and re-annotated the dataset with properly sized, tight bounding boxes around just the hand, which noticeably helped downstream training results.

Training Challenges

Early runs looked good on paper (mAP50 above 0.90) but performed poorly live on webcam. That gap between validation metrics and real-world behavior was one of the biggest lessons of the week: a validation set drawn from the same collection session as training data doesn't prove the model generalizes, it just proves it memorized that session. Over the course of the project I trained the model roughly 20 times, testing different combinations of augmented and non-augmented data, different epoch counts, and different image sizes. Surprisingly, the augmented runs frequently scored lower than the non-augmented ones on the same dataset — not just once, but repeatedly across multiple attempts, which pushed me to rely much more on real data collection than synthetic augmentation going forward.

Performance Improvements

- Added bounding-box size sanity filtering, higher inference resolution, and temporal smoothing across frames to reduce live false positives without needing new data.
- Compared YOLOv8n, YOLOv8s, and YOLOv8m on the same dataset and training setup — the smallest model (YOLOv8n) came out ahead on accuracy, size, and speed, showing the bottleneck was data, not model capacity.

Lessons Learned

One of the clearest lessons was around negative/background images — photos with no gesture in them at all. Before adding these, the model would sometimes draw confident bounding boxes on faces or background objects that weren't gestures at all, because it had never been shown an example of "nothing to detect here." Once negative images were added to the dataset, the model became noticeably more conservative and stopped firing on non-gesture content nearly as often. It was a small addition to the dataset that had a disproportionately large effect on real-world reliability, and it taught me that what a dataset is missing can matter just as much as what it contains.

The biggest shift in how I think about this project: a high mAP number is not the same as a good model. Real-world testing on webcam caught problems — false positives on faces, class confusion, scale mismatches — that the validation metrics never revealed. Diagnosing with a confusion matrix, rather than guessing from the outside, was the difference between wasting time on the wrong fix (more augmentation) and finding the right one (removing an ambiguous class, adding negative examples). I also learned that dropping a feature (the "call me" gesture) can be the more honest, evidence-backed engineering decision, rather than a failure — as long as it's documented with the reasoning behind it.

Week 3 Goals

- Approach Week 3 the same way I approached Week 2 — treat it as a chance to learn new methods and tools I haven't used before, not just repeat the same workflow.
- Spend more time researching techniques and best practices before jumping into implementation, since a lot of this week's progress came from fixing avoidable mistakes after the fact.

- Build on what worked well this week — targeted data collection, checking real-world performance early and often, and comparing multiple configurations side by side — and apply that same discipline to whatever Week 3's challenge turns out to be.
- Continue strengthening my understanding of the full computer vision pipeline so I can move faster and more confidently through dataset prep, training, and evaluation next time.